# EXPLORATORY DATA ANALYSIS (EDA) REPORT

**Insider Threat Detection Using Elastic Net and Random Forest**

## 1. Abstract

Insider threats are security risks caused by authorized users who intentionally or unintentionally perform activities that may compromise organizational information. This project focuses on detecting suspicious user behavior using machine learning. The Exploratory Data Analysis (EDA) stage is performed before model training to understand the dataset, identify important behavioral attributes, detect missing or duplicate records, examine distributions and outliers, study correlations, and understand the target-class distribution. The analyzed behavioral features are then prepared for Elastic Net and Random Forest models. Elastic Net provides regularization and useful feature selection, while Random Forest captures nonlinear relationships among behavioral features.

## 2. Introduction

Exploratory Data Analysis is a systematic process of examining a dataset before machine-learning model development. In insider-threat detection, EDA is particularly important because user activity can vary greatly between individuals and suspicious behavior may appear as unusual combinations or levels of otherwise normal activities.

The project analyzes activity such as logins, file access, email communication, device usage, and web activity. These raw activities can be transformed into user-level or user-period behavioral features suitable for classification.

## 3. Objectives of EDA

• Understand the structure and organization of the dataset.

• Identify numerical, categorical, and date/time attributes.

• Check missing values and duplicate records.

• Analyze descriptive statistics and feature distributions.

• Detect unusual values and potential outliers.

• Study relationships between behavioral features.

• Examine the distribution of normal and suspicious activity.

• Identify useful features for machine-learning models.

• Prepare the dataset for feature engineering and model training.

## 4. Dataset Description

The project is designed around the CERT Insider Threat Dataset, a simulated dataset containing different types of enterprise user activity. The raw data is organized around activity sources such as logon, device, file, email, HTTP/web, LDAP, and user-related information.

**Note:** The exact row count, column count, class counts, and missing-value statistics must be taken from the final dataset actually used in the implementation. They are intentionally not fabricated in this report.

| Activity Source | Example Information |
|---|---|
| Logon | Login and logout activity |
| Device | USB/device connection activity |
| File | File access and modification activity |
| Email | Email communication and attachments |
| HTTP/Web | Web browsing activity |
| LDAP/User | User and organizational information |
| Psychometric | Behavioral/personality-related information, where available |

# 5. Feature Description

| Feature | Purpose |
|---|---|
| login_count | Total login activity during the selected period |
| after_hours_logins | Login activity outside normal working hours |
| file_access_count | Number of file-access activities |
| file_copy_count | Number of file-copy activities, if available |
| usb_usage | Device/USB activity |
| email_count | Number of email activities |
| external_email_count | Email activity involving external recipients |
| email_attachment_count | Number of email attachments |
| web_activity_count | Number of web/HTTP activities |
| unique_domains | Number of distinct web domains visited |
| failed_logins | Unsuccessful login attempts, if available |
| after_hours_activity | Total activity occurring outside normal hours |
| department / role | Organizational context, where available |
| threat | Target class indicating normal or suspicious behavior |

## 6. Data Loading and Initial Inspection

The dataset is loaded into a Pandas DataFrame. The first inspection should include the first few records, dataset dimensions, column names, data types, and basic information about non-null values.

**Typical Python commands:**

```
df.head()
df.shape
df.columns
df.info()
df.describe()
```

The result of this step establishes the structure of the dataset and helps determine which preprocessing operations are required.

## 7. Missing Value Analysis

Missing values are checked because incomplete records can affect feature calculations and model training. Missing values should be quantified for every feature using df.isnull().sum(). Numerical values may be imputed using an appropriate statistical method, while categorical values can be handled using the most suitable category or an explicit unknown label. Columns with excessive or irrelevant missing information may be removed after inspection.

## 8. Duplicate Record Analysis

Duplicate records can cause certain activities to be counted more than once and may bias the machine-learning model. Duplicate records should therefore be identified using df.duplicated().sum() and removed where they represent genuine duplicate events.

## 9. Data Type Analysis

The dataset may contain numerical, categorical, and timestamp attributes. Numerical features can be used directly after appropriate scaling or transformation. Categorical features require encoding before model training. Timestamp information can be converted into meaningful behavioral features such as hour of day, day of week, or after-hours activity.

## 10. Descriptive Statistical Analysis

Descriptive statistics are obtained using df.describe(). Measures such as mean, standard deviation, minimum, maximum, median, and quartiles help understand the central tendency and spread of behavioral variables. Strongly skewed activity counts may require transformation or careful interpretation.

## 11. Univariate Analysis

Univariate analysis studies one feature at a time. Histograms and bar charts can be used to analyze login counts, file activity, email activity, USB usage, and web activity. This step helps identify skewness, concentration of values, and unusual activity ranges.

## 12. Outlier Analysis

Box plots can be used to identify unusually high or low activity. In this project, an extreme value is not automatically treated as an error because unusual behavior may itself contain information useful for

insider-threat detection. Outliers should therefore be investigated before deciding whether to remove, transform, or retain them.

## 13. Correlation Analysis

Correlation analysis examines relationships among numerical features. A correlation heatmap can reveal redundant or strongly related variables. For example, email count may be related to attachment count, while after-hours activity may be related to after-hours login activity. Correlation analysis supports feature selection and interpretation but does not by itself establish causation.

## 14. Class Distribution Analysis

The target variable should be examined using a frequency table and bar chart. Insider-threat datasets commonly contain substantially more normal activity than malicious or suspicious activity. If the project dataset is imbalanced, evaluation should not rely only on accuracy. Precision, recall, F1-score, confusion matrix, ROC-AUC, and where appropriate PR-AUC should also be considered.

## 15. Feature vs Target Analysis

Behavioral features can be compared across target classes using box plots, grouped bar charts, or other suitable visualizations. Useful comparisons include file access versus threat class, after-hours activity versus threat class, external email activity versus threat class, and USB activity versus threat class. These visualizations help identify features that may provide predictive information.

## 16. EDA Findings and Interpretation

1. User activity is heterogeneous; different users can have substantially different activity patterns.

2. Activity-count features may be skewed because a small number of users can generate many events.

3. After-hours activity can provide useful behavioral information.

4. File, email, device, login, and web activities can contribute to a user-behavior profile.

5. Potential class imbalance must be checked before model training.

6. Highly correlated features should be reviewed to reduce unnecessary redundancy.

7. Unusual values should be investigated rather than automatically deleted because anomalies can be relevant to the project objective.

8. Raw event data should be converted into meaningful aggregated features before classification.

## 17. EDA-Based Preprocessing Plan

| Step | Action |
|---|---|
| Data cleaning | Remove invalid and genuine duplicate records |
| Missing values | Impute or handle according to feature type and missingness |
| Time processing | Extract hour/day/after-hours behavioral indicators |
| Categorical encoding | Encode categorical variables where required |

| Step | Action |
| --- | --- |
| Feature aggregation | Aggregate raw events into user-level or user-period features |
| Scaling | Scale numerical features where appropriate, especially for Elastic Net |
| Feature selection | Review correlations and predictive usefulness |
| Train/test split | Separate training and testing data without leakage |

## 18. Relationship of EDA to the Proposed ML Models

**Elastic Net:** Elastic Net combines L1 and L2 regularization. It is useful when there are many behavioral features and when feature selection and regularization are important. EDA helps identify scale differences, redundant variables, missing values, and potentially useful predictors before Elastic Net training.

**Random Forest:** Random Forest is an ensemble of decision trees and can model nonlinear relationships and feature interactions. EDA helps identify meaningful behavioral variables and understand the distributions and target relationships before training the forest.

## 19. Recommended EDA Visualizations

• Dataset structure / feature summary table

• Missing-value summary

• Target-class distribution bar chart

• Histograms of major numerical features

• Box plots for major activity-count features

• Correlation heatmap

• Feature-versus-target box plots

• Activity comparison charts for normal versus suspicious users

## 20. Conclusion

EDA provides the foundation for the insider-threat detection pipeline. It allows the project team to understand user behavior, detect data-quality issues, identify useful features, examine class imbalance, and prepare the dataset for machine learning. The final cleaned behavioral dataset can then be supplied to Elastic Net and Random Forest models for insider-threat classification and evaluation.

## 21. Important Implementation Note

This report is a submission-ready EDA report structure for the project, but exact dataset statistics such as total rows, total columns, missing-value counts, duplicate counts, and class percentages must be generated from the actual dataset file used by the project. Replacing these with guessed numbers would make the report inaccurate.